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## ELECTROMAGNETIC INTERFERENCE SHIELDING STRUCTURES

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### Abstract

Embodiments of the present invention relate to helical electromagnetic interference shielding (“HES”) structures. In one general aspect, the HES structure includes a folded state, unfolded state, and pair of rectangular EMI shielding panels. The EMI shielding panels form interconnected EMI shielding planes. Each rectangular EMI shielding panel includes a substrate and conductive layer. The conductive layer is positioned near the substrate and includes a conductive composition. The conductive composition includes graphene sheets dispersed in a polymer matrix as a 3D percolated network. The substrate is rectangular and includes a film and/or fabric. The EMI shielding planes are angularly offset from each other about their center axis and form a helical structure when in the unfolded state. Each EMI shielding plane rotates about the center axis when the HES structure transitions between the folded and unfolded states. Adjacent EMI shielding planes are an angular offset from each other.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/241,689 filed Sep. 1, 2023, which is a continuation of U.S. patent application Ser. No. 17/986,529 filed Nov. 14, 2022 (issued as U.S. Pat. No. 11,882,680 on Jan. 3, 2024), which claims priority to U.S. Provisional Application No. 63/247,753 filed Sep. 23, 2021, the contents of which are incorporated by reference herein in their entirety.

### BACKGROUND

[0002] The present invention relates generally to electromagnetic interference (“EMI”) and specifically to EMI shielding structures. All electronics emit some level of radio frequency known as “electromagnetic radiation.” EMI occurs in the radio frequency spectrum when there is a disturbance generated by an external source that affects an electrical circuit by electromagnetic induction, electrostatic coupling, or conduction. In communication devices (e.g., cell phones, computers, Bluetooth devices, laptops), commercial appliances (i.e., microwave ovens, the design of microwave circuits) and the automotive industries (i.e., integrated electrical circuits), EMI deteriorates the durability and proper functioning of electronic equipment.

[0003] EMI shielding is defined in terms of the reflection and/or absorption of EM radiation by a material that acts as a barrier against the penetration of the radiation passing through the shielding materials. These materials prevent the transmission of EM radiation by reflection and/or absorption of the EM radiation or by suppressing the EM signals so that EM waves do not affect the functioning and durability of electronic equipment. There exists a need in the space for three-dimensional structures that absorb and/or reflect EMI radiation.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 illustrates a side view of a hexagonal electromagnetic interference shielding (“HES”) structure in an unfolded state, generally **100**, in accordance with some embodiments.

[0005] FIG. 2 illustrates a top view EMI shielding panels orthogonally positioned at their first end and their method successively folding over each other to form the HEX structure **100**, in accordance with other embodiments.

[0006] FIG. 3 illustrates a perspective view of an EMI shielding panel, generally **200**, and View A, according to certain embodiments.

[0007] FIG. 4 illustrates View A, which depicts a cross-section of EMI shielding panel **200** that reflects a multilayered structure, in accordance with yet still other embodiments.

[0008] FIG. 5 illustrates a side view of the HES structure **100** in a folded state, in accordance with yet still some embodiments.

[0009] FIG. **6** illustrates a top view of the HES structure **100** in the folded state, in accordance with other embodiments.

[0010] FIG. **7** illustrates a top view of the HES structure **100** in the unfolded state, in accordance with certain embodiments.

[0011] FIG. **8** illustrates a top view of the angular offset, generally **805**, of two interconnected EMI shielding planes, generally **105a** and **105b**, in accordance with yet still other embodiments.

[0012] FIG. **9** illustrates a side view of an EMI shielding construct, generally **900**, in a folded state, in accordance with some embodiments.

[0013] FIG. **10** illustrates a side view of the EMI shielding construct **900**, in an unfolded state, in accordance with other embodiments.

[0014] FIG. **11** illustrates a top view of an EMI shielding construct, generally **1100**, in accordance with certain embodiments.

[0015] FIG. **12** illustrates an EMI shielding assembly, generally **1200**, that includes a plurality of HES structures **100** arranged in rows and columns with an  $L/2$  spacing, in accordance with yet still other embodiments.

[0016] FIG. **13** illustrates an EMI shielding assembly, generally **1300**, that includes a plurality of HES structures **100** arranged in rows and columns with an  $L/3$  spacing, in accordance with some embodiments.

[0017] FIG. **14** illustrates an EMI shielding structure, generally **1400**, that includes a plurality of EMI shielding constructs, generally **1405**, arranged in a parallelogram, in accordance with other embodiments.

[0018] FIG. **15** illustrates an EMI shielding structure, generally **1500**, that includes a plurality of EMI shielding constructs, generally **1505**, arranged in a septagon, in accordance with certain embodiments.

[0019] FIG. **16** depicts the method steps to form an EMI structure, in accordance with yet still other embodiments.

[0020] FIG. **17** depicts additional method steps to form an EMI structure, in accordance with some embodiments.

[0021] FIG. **18** depicts additional method steps to form an EMI structure, in accordance with other embodiments.

[0022] FIG. **19** depicts additional method steps to form an EMI structure, in accordance with certain embodiments.

[0023] FIG. **20** depicts additional method steps to form an EMI structure, in accordance with yet still other embodiments.

#### DETAILED DESCRIPTION

[0024] The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

[0025] Certain terminology may be employed in the following description for convenience rather than for any limiting purpose. For example, the terms “forward” and “rearward,” “front” and “rear,” “right” and “left,” “upper” and “lower,” and “top” and “bottom” designate directions in the drawings to which reference is made, with the terms “inward,” “inner,” “interior,” or “inboard” and “outward,” “outer,” “exterior,” or “outboard” referring, respectively, to directions toward and away from the center of the referenced element, the terms “radial” or “horizontal” and “axial” or “vertical” referring, respectively, to directions or planes which are perpendicular, in the case of radial or horizontal, or parallel, in the case of axial or vertical, to the longitudinal central axis of the

referenced element, and the terms “downstream” and “upstream” referring, respectively, to directions in and opposite that of fluid flow. Terminology of similar import other than the words specifically mentioned above likewise is to be considered as being used for purposes of convenience rather than in any limiting sense.

[0026] In the figures, elements having an alphanumeric designation may be referenced herein collectively or in the alternative, as will be apparent from context, by the numeric portion of the designation only. Further, the constituent parts of various elements in the figures may be designated with separate reference numerals which shall be understood to refer to that constituent part of the element and not the element as a whole. General references, along with references to spaces, surfaces, dimensions, and extents, may be designated with arrows. Angles may be designated as “included” as measured relative to surfaces or axes of an element and as defining a space bounded internally within such element therebetween, or otherwise without such designation as being measured relative to surfaces or axes of an element and as defining a space bounded externally by or outside of such element therebetween. Generally, the measures of the angles stated are as determined relative to a common axis, which axis may be transposed in the figures for purposes of convenience in projecting the vertex of an angle defined between the axis and a surface which otherwise does not extend to the axis. The term “axis” may refer to a line or to a transverse plane through such line as will be apparent from context.

[0027] The present invention relates generally to electromagnetic interference (“EMI”) and specifically to EMI shielding structures. All electronics emit some level of radio frequency known as “electromagnetic radiation”. EMI occurs in the radio frequency spectrum when there is a disturbance generated by an external source that affects an electrical circuit by electromagnetic induction, electrostatic coupling, or conduction. In communication devices (e.g., cell phones, computers, Bluetooth devices, laptops), commercial appliances (i.e., microwave ovens, the design of microwave circuits) and the automotive industries (i.e., integrated electrical circuits), EMI deteriorates the durability and proper functioning of electronic equipment.

[0028] A number of solutions have been developed over the years to combat these undesired signals. For example, there are two types of EMI shielding: conductive coatings and EMI mesh. Transparent conductive coatings typically offer less than 40 dB EMI emission attenuation with a sheet resistance from 1 to 100 Q/sq. EMI mesh (e.g., wire mesh, micromesh, and laser mesh) usually offers 60-80 dB of EMI emission attenuation and conductivity of less than 0.3 Q/sq. Each option offers varying levels of light transmission, attenuation, and sheet resistance. EMI filters can be tuned for a specific frequency. For example, the effectiveness of an EMI shield can be determined by testing the level of the signal passing through the aperture (measured in decibels “dB”) before and after fitting the shield then comparing the difference.

[0029] EMI shielding is defined in terms of the reflection and/or absorption of EM radiation by a material that acts as a barrier against the penetration of the radiation passing through the shielding materials. These materials prevent the transmission of EM radiation by reflection and/or absorption of the EM radiation or by suppressing the EM signals so that EM waves do not affect the functioning and durability of electronic equipment. In general, conductive materials like metals, owing to their high reflectivity, are widely used to isolate spaces or equipment from surrounding EM waves. This reflection shielding is based on the principle of the Faraday cage, in which inside the cage, space is completely impervious to external electric fields.

[0030] On the other hand, absorption shielding is related to permeable materials (i.e., magnetic materials). Accordingly, metallic conductors suffer a lack of flexibility, heaviness, and high costs. Meanwhile, ferromagnetic or ferrimagnetic materials have an intrinsic cut-off frequency, usually below the low GHz range, that hinders their use in EMI shielding over a broad GHz range. There exists a need in the space for three-dimensional (“3D”) structures that absorb and/or reflect EMI radiation over a broad GHz range that is flexible, light weight, low cost, and is capable of 60-80 dB of EMI emission attenuation and conductivity of less than 0.3 Q/sq.

[0031] Embodiments of the instant disclosure seek to provide EMI shielding structures that can be assembled in a variety of 3D EMI shielding constructs (e.g., walls, curtains, dividers, panels, liners, barriers, etc.) that can be used, for example, to shield rooms, buildings, sheds, tents, storage containers, as well as other 3D structures. The 3D EMI shielding constructs of the instant disclosure can also be assembled into 3D EMI shielding structures (e.g., rooms, building, sheds, as well as other structures having a roof and walls). Other aspects of the instant disclosure seek to provide multifaceted EMI shielding structures that utilize various layers of metals, conductive inks, and fabrics. The instant disclosure further seeks to provide helical EMI shielding structures that have a folded state and an unfolded state.

[0032] Shielding efficiency (SET) is typically defined as a parameter that measures how well a material impedes the EM energy of a certain frequency when passing through it. When the EM waves fall on the front-face of the shielding material then a certain part of the incident power (Pr) is reflected (PR), while a certain part is absorbed and dissipated in form of energy, and the remaining part is transmitted (Pr) through the shielding material. Therefore, three different processes namely reflection, absorption and multiple internal reflections contribute to the whole attenuation, corresponding to shielding effectiveness SE.sub.R, SE.sub.A and SE.sub.M, respectively ([EQ. 1]).

$$[00001] \text{SE}_T = 10\log\frac{P_1}{P_T} = 20\log\frac{E_1}{E_T} = 20\log\frac{H_1}{H_T} = \text{SE}_R + \text{SE}_A + \text{SE}_m \quad [\text{EQ. 1}]$$

[0033] Here P, E and H refer to power and electric and magnetic field intensities while subscripts I, R and T represent the incident, reflected and transmitted components, respectively. Thus, SE.sub.R refers to net reflection and SE.sub.A represents shielding due to absorption. The primary mechanism of EMI shielding is reflection. Reflection loss (SE.sub.R) is related to the relative impedance mismatching between the surface of the shielding material and the EM waves. SE.sub.R is a function of the ratio of conductivity and permeability of the material. Therefore, materials must have mobile charge carriers (electrons or holes) for reflection of the EM radiation.

[0034] A secondary mechanism of EMI shielding is absorption. As obtained from the plane wave theory, the amplitude of the EM wave decreases exponentially inside the material as it passes through it. Absorption loss results from ohmic losses and heating of the material due to the currents induced in the medium. SE.sub.A is directly related to conductivity, permeability and sample thickness (i.e., EMI absorption is proportional to sample thickness). Such a dependency of SE.sub.R/SE.sub.A on  $\mu$  (permeability) and  $\sigma$  (conductivity) indicates that in magnetic conducting metals, shielding is dominated by absorption rather than reflection. For thinner materials, radiation is trapped between two boundaries due to multiple reflection (i.e., EM waves reflect from the second boundary, come back to first boundary and are re-reflected from the first to second boundary, and so on).

[0035] Multiple reflection plays an important role for porous structures and some definite geometries. Here, a large surface area and a big vacant space excluding the solid structure gives more active sites for scattering and multiple reflection of electromagnetic waves. The hollow/porous structure shows unique properties, e.g., high surface area, disciplinable internal structures, low density and complimentary permeability that can fulfill the quest for improving EMI shielding performance. These multiple reflections (SE.sub.M) can be neglected when the thickness of the shielding materials is greater than the penetration depth or when SE.sub.A is more than 10 dB because in thick shielding materials (high SE.sub.A) the EM wave hits at the second boundary with negligible amplitude so SE.sub.M can be neglected.

[0036] Therefore, there is a complex dependency on the geometry, size, conductivity and volume fraction of each constituent. For example, the essential requirement is to adjust the effective permittivity and permeability to certain values by which reflection can be minimized. Therefore, a prerequisite of conductive EMI shielding composites is to limit reflection and enhance absorption for effective EMI shielding materials.

[0037] Given the important role electrical conductivity plays in shielding effectiveness, shielding

materials are typically made of metals such as copper, aluminum, brass, nickel, silver, steel, and tin in the form of foils, screens, films, braids, foams, and enclosures. Although effective, the use of metals alone as EMI shielding material possess many disadvantages compared to solutions that incorporate conductive compositions of the instant disclosure (e.g., greater weight, higher costs, reduced flexibility, and increased susceptibility to corrosion). The EMI shielding structures of the instant disclosure utilize a conductive composition that includes a polymer(s) and fully exfoliated single sheets of graphene and provides EMI shielding without the drawbacks of using metals alone.

[0038] Turning now to the Figures. FIG. 1 illustrates a side view of a hexagonal electromagnetic interference shielding (“HES”) structure in an unfolded state **130**, generally **100**, in accordance with some embodiments. The HES structure **100** includes a plurality of interconnected EMI shielding planes **105** (e.g., EMI shielding panels **105a** is interconnected to EMI shielding panels **105b**). Specifically, each plurality of interconnected EMI shielding planes **105** includes a pair of rectangular EMI shielding panels **200**. The HES structure **100** includes a top end **110** and an oppositely positioned bottom end **115**. The HES structure **100** has a helical structure **125** due to the manner in which the interconnected EMI shielding planes **105** are connected and positioned relative to each other.

[0039] FIG. 2 illustrates a top view of the EMI shielding panels **200** orthogonally positioned at their first end **205** and their method of successively folding over each other to form the HEX structure **100**, in accordance with other embodiments. Each EMI shielding panel **200** includes a first end **205** and a second end **210**. To form an HES structure **100**, a pair of EMI shielding panels **200** are affixed orthogonal to each other at their first ends **205** and successively overlapped with each other to thereby form the plurality of interconnected EMI shielding planes **105**. Subsequently, the EMI shielding panels **200** are affixed at their second ends **210** to thereby form the HES structure **100**, which unfolds to an unfolded state **130** (depicted in FIG. 1) and folds to a folded state **505** (depicted in FIG. 5) along its center axis **120**. As depicted in FIG. 1, the plurality of interconnected EMI shielding planes **105** are each angularly offset from each other about the center axis **120** to form the helical structure **125** when in the unfolded state **130**.

[0040] FIG. 3 illustrates a perspective view of the EMI shielding panel **200** and View A, according to certain embodiments. The EMI shielding panel **200** is preferably multilayered to not only provide EMI shielding but also provide structural as well as environmental integrity. For example, since the HES structure **100** includes folded EMI shielding panels **200**, the HES structures must have reduced corrosion susceptibility compared to metal EMI shielding structures and in light of their inclusion in 3D constructs and 3D structures (discussed further below) they preferably should be able to form and hold predetermined contours.

[0041] FIG. 4 illustrates View A, which depicts a cross-section of the EMI shielding panel **200** that reflects a multilayered structure (e.g., a laminate), in accordance with yet still other embodiments. The EMI shielding panel **200** preferably includes a substrate **405**, at least one conductive layer **410** positioned proximate to on one or more sides (e.g., a side **406** and/or oppositely positioned side **407**) of the substrate **405**. Depending on which layers are present, an encapsulating layer **415** can be positioned proximate to the conductive layer **410**, the substrate **405**, and/or the conductive layer **410**. In other words, the encapsulating layer **415** encapsulates the layers that are present EMI shielding panel **200**. In some embodiments, a metallic substrate **420** can be proximate to the conductive layer **410**, the substrate **405**, the conductive layer **410**, and/or the encapsulating layer **415**.

[0042] View A depicts one of many configurations that the EMI shielding panel **200** can assume. In preferred embodiments, the EMI shielding panel **200** is a laminate that includes at least one substrate and a plurality of layers. Here, the substrate **405** includes the conductive layer **410** positioned proximate to the first side **406** and the second side **407**. The substrate **405** can include films (e.g., PET, polyethylene, polyetherimides, polyurethanes, thermoplastic polyurethanes, and other polymeric films) as well as woven and/or unwoven fabrics (e.g., nylons, cotton, rayon,

metallic fabrics).

[0043] An encapsulating layer **415** is positioned proximate to each of the conductive layers **410** to thereby form an encapsulating body **425**. A metallic substrate **420** is positioned proximate to each side of the encapsulating body **425** (i.e., proximate to each of the encapsulating layers **415**) and thereby forms a shrouded body **430**. The shrouded body **430** is subsequently rectangularly parsed to form the EMI shielding panels **200**. In embodiments that lack the metallic substrate **420**, the encapsulating body is parsed to form the EMI shielding panels **200**.

[0044] The conductive composition in the form of the conductive layer **410** can be applied as a coating or a preformed layer(s). When applied as a coating, the conductive layer **410** is preferably cured before that addition of subsequent layers. For example, the conductive layer **410** can be applied using one or more coating methods known in the art (e.g., screen printing, vapor deposition, electrochemical techniques, spraying, roll-to-roll coating, physical coating, and/or other coating techniques known in the art). The conductive layer **410** can include one or more layers of the conductive composition where each layer can be up to 80 nm thick. The conductive layer **410** can be pre-formed using a mold, a cast, a punch, and/or similar manufacturing techniques. Such pre-formed conductive layers **410** can subsequently be cured. The substrate **405** is preferably planar, flexible, and includes a film and/or fabric.

[0045] Preferably, the conductive composition (including applicable polymers and methods of manufacture) as well as the fully exfoliated single sheets of graphene can be prepared as disclosed in U.S. Pat. No. 7,658,901 B2 to Prud'Homme et al., U.S. Pat. No. 8,679,485 B2 to Crain et al., U.S. Pat. No. 8,278,757 B2 to Crain et al., and U.S. Patent Publication No. 2011/0189452 A1 to Lettow et al., which are each herein incorporated in their entirety. The graphene sheets preferably have a maximum surface area of 2630 m<sup>2</sup>/g, in accordance with certain embodiments. In several embodiments, the graphene sheets are present in the polymer as a three-dimensional percolated network (e.g., a continuous three-dimensional network comprising continuous chains of graphene sheets).

[0046] Applicable polymers include, but are not limited to, silicone, polyurethane, polyester, polyethylene, polypropylene, polystyrene, polyurea, polyimide, polyamide, polyethylene terephthalate, polycarbonate, polyvinylchloride, polyether ether ketone, acrylonitrile butadiene styrene, polylactic acid, methacrylate, epoxy, phenolics, rubber, and similar polymers. In certain embodiments, conductive layers **410** may be formed by one or more currently available methods, such as extruding, calendaring, and pressing.

[0047] In some embodiments, the conductive composition can further include one or more electromagnetic absorbing material (e.g., carbon black, graphite, graphite oxide, expanded graphite, carbon fibers, carbon nanotubes, etc.) uniformly dispersed within the polymer matrix. The conductive composition includes fully exfoliated graphene sheets dispersed in one or more polymers and may include additional components such as emulsifiers, solvents, dispersion aids, etc. In other embodiments, the three-dimensional percolated network comprises a graphene sheet network comprising nanometer scale separation at the contact points between individual sheets. In yet still other embodiments, individual graphene sheets may comprise imperfections in its lattice network (i.e., kinks) that facilitate the interlocking of individual graphene sheets in the percolated network.

[0048] Structurally, graphene has hybrid orbitals formed by sp<sup>2</sup> hybridization. In the sp<sup>2</sup> hybridization, the 2s orbital and two of the three 2p orbitals mix to form three sp<sup>2</sup> orbitals. The one remaining p-orbital forms a pi-bond between the carbon atoms. Similar to the structure of benzene, the structure of graphene has a conjugated ring of the p-orbitals which exhibits a stabilization that is stronger than would be expected by the stabilization of conjugation alone, i.e., the graphene structure is aromatic. Unlike other allotropes of carbon such as diamond, amorphous carbon, carbon nanofoam, or fullerenes, graphene is not an allotrope of carbon since the thickness of graphene is one atomic carbon layer i.e., a sheet of graphene does not form a three-dimensional

crystal.

[0049] Graphene has an unusual band structure in which conical electron and hole pockets meet only at the K-points of the Brillouin zone in momentum space. The energy of the charge carriers, i.e., electrons or holes, has a linear dependence on the momentum of the carriers. As a consequence, the carriers behave as relativistic Dirac-Fermions having an effective mass of zero and moving at the effective speed of light of 106 msec. Their relativistic quantum mechanical behavior is governed by Dirac's equation. As a consequence, graphene sheets have a large carrier mobility of up to 60,000 cm<sup>2</sup>/V-sec at 4K at 300K, the carrier mobility is about 15,000 cm<sup>2</sup>/V-sec. Also, quantum Hall effect has been observed in graphene sheets.

[0050] The linear dispersion of graphene around the K (K') point leads to constant interband absorption (from valence to conduction bands, about 2.3%) of vertical incidence light in a very broadband wavelength range. More interestingly, at the microwave and terahertz frequency ranges, intraband absorption dominates and a single layer can absorb as much as 30% at a light wavelength of 300 microns depending on the carrier concentration in the. As a result, utilization of graphene for microwave and terahertz frequency absorption has numerous advantages such as being an ultra-thin and efficient absorption layer relative to other materials.

[0051] Moreover, because graphene is a one atom thick monolayer sheet formed of carbon atoms packed in a honeycomb crystalline lattice, wherein each carbon atom is bonded to three adjacent carbon atoms via sp<sup>2</sup> bonding, the overall thickness required to provide effective absorption is minimal on the order of a few nanometers. As such, the use of graphene sheets provides minimal added weight to the object to be shielded, has broadband absorption capabilities, and provides greater versatility than prior art structures. Moreover, graphene is generally recognized for its high mechanical strength and high stability which are desirable properties for most applications.

[0052] Examples of polymeric materials include, but are not limited to, those comprising thermoplastics and thermosets, including elastomers and rubbers (including thermoplastics and thermosets), silicones, fluorinated polysiloxanes, natural rubber, butyl rubber, chlorosulfonated polyethylene, chlorinated polyethylene, styrene/butadiene copolymers (SBR), styrene/ethylene/butadiene/styrene copolymers (SEBS), styrene/ethylene/butadiene/styrene copolymers grafted with maleic anhydride, styrene/isoprene/styrene copolymers (SIS), polyisoprene, nitrile rubbers, hydrogenated nitrile rubbers, neoprene, ethylene/propylene copolymers (EPR), ethylene/propylene/diene copolymers (EPDM), ethylene/vinyl acetate copolymer (EVA), hexafluoropropylene/vinylidene fluoride/tetrafluoroethylene copolymers, tetrafluoroethylene/propylene copolymers, fluorelastomers, polyesters (such as poly(ethylene terephthalate), poly(butylene terephthalate), poly(ethylene naphthalate), liquid crystalline polyesters, poly(lactic acid), etc.); polystyrene; polyamides (including polyterephthalamides); polyimides (such as Kapton®); aramids (such as Kevlar® and Nomex®); fluoropolymers (such as fluorinated ethylene propylene (FEP), polytetrafluoroethylene (PTFE), poly(vinyl fluoride), poly(vinylidene fluoride), etc.); polyetherimides; poly(vinyl chloride); poly(vinylidene chloride); polyurethanes (such as thermoplastic polyurethanes (TPU); spandex, cellulosic polymers (such as nitrocellulose, cellulose acetate, etc.); styrene/acrylonitriles polymers (SAN); acrylonitrile/butadiene/styrene polymers (ABS); polycarbonates; polyacrylates; poly(methyl methacrylate); ethylene/vinyl acetate copolymers; thermoset epoxies and polyurethanes; polyolefins (such as polyethylene (including low density polyethylene, high density polyethylene, ultrahigh molecular weight polyethylene, etc.), polypropylene (such as biaxially-oriented polypropylene, etc.); Mylar; etc.

[0053] The substrate can be non-woven materials, such as DuPont Tyvek®. The substrate can be an adhesive or adhesive-backed material (such as adhesive-backed papers or paper substitutes). The substrate can be mineral-based paper substitutes such as Teslin® from PPG Industries, Inc. The substrate can be a transparent or translucent or optical material, such as glass, quartz, polymer (such as polycarbonate or poly(meth)acrylates (such as poly(methyl methacrylate).



[0054] For example, components of the nanocomposites disclosed herein, such as two or more of the functionalized graphene sheets, binders, carriers, and/or other components may be blended by using suitable mixing, dispersing, and/or compounding techniques and apparatus, including ultrasonic devices, high-shear mixers, two-roll mills, three-roll mills, cryogenic grinding crushers, extruders, kneaders, double planetary mixers, triple planetary mixers, high pressure homogenizers, ball mills, attrition equipment, sand mills, and horizontal and vertical wet grinding mills, and the like.

[0055] By one approach, the conductive layer **410** can be formed using one or more of the following methods, including, but not limited to, painting, coating, printing, pouring, spin casting, solution casting, dip coating, powder coating, by syringe or pipette, spray coating, curtain coating, lamination, co-extrusion, spray coating, electrospray deposition, ink-jet printing, spin coating, thermal transfer (including laser transfer) methods, screen printing, rotary screen printing, gravure printing, capillary printing, offset printing, electrohydrodynamic (EHD) printing (as disclosed in WO 2007/053621, which is herein incorporated by reference in its entirety), flexographic printing, pad printing, stamping, microcontact printing, dip pen nanolithography, laser printing, via pen or similar means, and the like.

[0056] After one or more layers are formed, the layer may be cured using any suitable technique, including drying and oven-drying (in air or another inert or reactive atmosphere), UV curing, IR curing, microwave curing or drying, and the like.

[0057] FIG. **5** illustrates a side view of the HES structure **100** in a folded state, in accordance with yet still some embodiments. In the folded state, the plurality of interconnected EMI shielding planes **105** are aligned with each other, as illustrated in the top view of the HES structure **100** reflected in FIG. **6**. As illustrated in the top view reflected in FIG. **7**, the plurality of interconnected EMI shielding planes **105** each axially rotate about the center axis **120** when the HES structure **100** transitions between the folded state and unfolded state.

[0058] As illustrated in FIG. **8**, the angular offset **805** between adjacent EMI shielding planes **105** (e.g., EMI shielding plane **105a** and EMI shielding plane **105b**) increases as the distance between the adjacent EMI shielding structures increases and decreases when the distance decreases. In other words, the angular offset **805** increases as the HES structure **100** transitions from the folded state to the unfolded state and decreases when transitioning from the unfolded state to the folded state.

[0059] In other embodiments, the HES structures **100** can be arranged in a plurality of rows and columns to form an EMI shielding construct that has a 3D geometric shape. The EMI shielding constructs can be in the form of blocks, walls, dividers, and/or panels that are, for example, used to form EMI shielding structures having a roof and/or walls. In other words, the thus formed EMI shielding structures provide an EMI shielded space that protects electronic devices and circuits from EMI radiation.

[0060] FIG. **9** illustrates a side view of an EMI shielding construct, generally **900**, in a folded state **920**, in accordance with some embodiments. The EMI shielding construct **900** includes a plurality of HES structures **100** (e.g., the HES structures **100a**, **100b**, and **100c**) that are rigidly affixed to a first rigid panel **910** via their first end **110** and rotatably affixed to a second rigid panel **915** via their second end **115**. Alternatively, the panel **910** and/or the panel **915** can be flexible. Alternatively, the first end **110** can be rotatably affixed and the second end **115** can be rigidly affixed. To be sure, one end of each HES structure **100** must be affixed to allow rotation about the central axis **120**.

[0061] FIG. **10** illustrates a side view of the EMI shielding construct **900**, in an unfolded state **1020**, in accordance with other embodiments. Here, the plurality of HES structures **100** are longitudinally positioned side by side in one or more rows and columns to thereby form the EMI shielding construct **900**. Although depicted as uniform in width, the plurality of HES structures **100** can have different widths to conform to any predetermined geometric configuration. The plurality of HES structures **100** are preferably arranged to eliminate gaps therebetween that may be present when in the unfolded state to thereby form a barrier that blocks and/or absorbs EMI radiation.

Turning now to alternative views and configurations of EMI constructs.

[0062] FIG. **11** illustrates a top view of an EMI shielding construct, generally **1100**, in accordance with certain embodiments. Here, the top view illustrates the plurality of HES structures **100** longitudinally arranged in rows and columns in the form of a parallelogram configuration to thereby form the EMI shielding construct **1100**. In some embodiments, the EMI shielding constructs (e.g., EMI shielding construct **1100**) contains an outer shell (e.g., a shell **1105**) that may provide structural integrity and/or additional EMI shielding protection.

[0063] The plurality of HES structures **100** are preferably spaced apart according to their width to minimize and/or eliminate the aforementioned potential gaps. Applicable spacing includes, but is not limited to, W/8-W/9, wherein W is the width of the HES structure **100**. FIG. **12** illustrates a top view of an EMI shielding assembly, generally **1200**, that includes a plurality of HES structures **100** arranged in rows **1205** and columns **1210** with a W/2 spacing, in accordance with yet still other embodiments. FIG. **13** illustrates a top view of an EMI shielding assembly, generally **1300**, that includes a plurality of HES structures **100** arranged in rows and columns with a W/3 spacing, in accordance with some embodiments.

[0064] FIG. **14** illustrates a top view of an EMI shielding structure, generally **1400**, that includes a plurality of EMI shielding constructs, generally **1405**, arranged in a parallelogram configuration, in accordance with other embodiments. The EMI shielding construct **1405** includes a shell **1410**. Although not depicted, the EMI shielding structure **1400** can include an EMI shielding roof (e.g., an EMI shielding panel horizontally positioned on the EMI shielding structure **1400**). FIG. **15** illustrates a top view of an EMI shielding structure, generally **1500**, that includes a plurality of EMI shielding constructs, generally **1505**, arranged in a septagonal configuration, in accordance with certain embodiments. The EMI shielding construct **1505** includes a shell **1510**. Although not depicted, the EMI shielding structure **1500** can include an EMI shielding roof (e.g., an EMI shielding panel horizontally positioned on the EMI shielding structure **1500**).

[0065] FIG. **16** depicts the method steps to form an EMI structure **100**, in accordance with yet still other embodiments. At Step **1605**, rectangular EMI shielding panels **200** are formed that each include a first end **205** and a second end **210**. At Step **1610**, a pair of rectangular EMI shielding panels **200** are affixed orthogonal to each other at their first ends **205** (e.g., see FIG. **2**). At Step **1615**, the pair of rectangular EMI shielding panels **200** are successively overlapped with each other to thereby form a plurality of interconnected EMI shielding planes **105** (e.g., see FIG. **5**).

[0066] At Step **1620**, the pair of rectangular EMI shielding panels **200** are affixed to each other at their second ends **210** (e.g., the second end **210a** and the second end **210b**) to thereby form a helical EMI shielding (“HES”) structure **100** that unfolds to an unfolded state **130** (e.g., see FIG. **1**) and folds to a folded state **505** (e.g., see FIG. **5**) along its center axis **120**. Here, the plurality of interconnected EMI shielding planes **105** are each angularly offset from each other (e.g., via angular offset **805**) about the center axis **120** to form a helical structure **125** when in the unfolded state **130**.

[0067] The encapsulating layer **410** includes PET, polyetherimides, PU, thermoplastic polyurethane (TPU), ABS, PVC, nylon, PLA, polybenzimidazole, polycarbonate, polyether sulfone, polyoxymethylene, polyether ether ketone, polyetherimide, polyethylenes, polypropylene, polystyrene, polyvinyl chloride, polyvinylidene fluoride, Teflon, and/or other thermoplastic films known in the art.

[0068] The metallic layer **420** is preferably in the form of a metallic film, metallic screen/mesh and/or a metallic fiber. Applicable metals include, but are not limited to, silver, gold, copper, aluminum, steel, zinc, nickel, titanium, and/or an alloy. To be sure, the metallic layer **420** provides additional structural integrity to the HES structure **100**. FIG. **17** depicts additional method steps to form the EMI structure of FIG. **16**, in accordance with some embodiments. At Step **1705**, a conductive composition is formed that includes a polymer and fully exfoliated single sheets of graphene that are present as a three-dimensional percolated network within the polymer. At Step

**1710**, to form the rectangular EMI shielding panels of Step **1605**, the conductive composition is applied to a side **406** (or a side **407**) of a substrate **405** to thereon form a conductive layer **410**. [0069] Here, the substrate **405** includes one or more of a film or a fabric. Applicable films include, but are not limited to, polymeric films such as those that include polyethylene terephthalate (PET), polyethylene, polyetherimides, polyurethanes (PU), and/or similar thermoplastics. Applicable fabrics include, but are not limited to, polymeric fabrics such as those that include nylon and/or rayon, natural fabrics as well as those that include silver, copper, nickel, aluminum, zinc, brass and/or steel. Such fabrics can be woven or non-woven. In certain embodiments, the substrate **405** is in the form of a mesh. At Step **1715**, the coated substrate **405** is parsed into rectangular panels **200**. [0070] FIG. **18** depicts additional method steps to form an EMI shielding structure **100**, in accordance with other embodiments. At Step **1805**, a plurality of HES structures **100** are rigidly affixed to a first rigid panel **910** via a top end **110**, the plurality of HES structures **100** longitudinally positioned side by side in one or more of a row **1205** or a column **1210**. At Step **1810**, the plurality of HES structures is rotatably affixed to a second rigid panel **915** via a bottom end **115**. [0071] FIG. **19** depicts additional method steps to form an EMI shielding structure **100**, in accordance with certain embodiments. At Step **1905**, a conductive composition is formed (e.g., using one or more of the aforementioned techniques) that includes a polymer and fully exfoliated single sheets of graphene that are present as a three-dimensional percolated network within the polymer. At Step **1910**, to form the rectangular EMI shielding panels **200** of Step **1605**, the conductive composition is applied on a side **406** (or a side **407**) of a substrate **405** to thereon form a conductive layer **410**, the substrate **405** is planar, flexible, and includes one or more of a first film and a first fabric. At Step **1915**, an encapsulating layer **415** is positioned proximate to one or more of the conductive layer **410** and the substrate **405** to thereby form an encapsulated body **425**. [0072] Here, the encapsulating layer **415** includes one or more of a film, a fabric, a polymer. At Step **1920**, the encapsulated body **425** is parsed into rectangular panels **200**. FIG. **20** depicts additional method steps to form an EMI shielding structure **100**, in accordance with yet still other embodiments. At Step **2005**, a conductive composition is formed (e.g., using one or more of the aforementioned techniques) that includes a polymer and fully exfoliated single sheets of graphene that are present as a three-dimensional percolated network within the polymer. At Step **2010**, to form the rectangular EMI shielding panels (Step **1605**), the conductive composition is applied on a side **406** (and/or a side **407**) of a substrate **405** to thereon form a conductive layer **410**. Here, the substrate **405** is preferably planar, flexible, and includes one or more of a film and a fabric. [0073] At Step **2010**, an encapsulating layer **415** is positioned proximate to one or more of the conductive layer **410** and the substrate **405** to thereby form an encapsulated body **425**. Here, the encapsulating layer **415** includes one or more of a film, a fabric, and a polymer. At Step **2015**, metallic layer **420** is applied to the encapsulated body **425** to thereby form a shrouded body **430**. Here, the metallic layer **420** is planar, flexible, and includes one or more of a metallic film, metallic fiber, and a metallic screen. At Step **2025**, the shrouded body is parsed into rectangular panels **200**. [0074] Based on the foregoing, a HES structure and associated methods of manufacture have been disclosed in accordance with the instant disclosure. However, numerous modifications and substitutions can be made without deviating from the scope of the instant disclosure. Therefore, the instant disclosure has been presented by way of example and not limitation.

## Claims

1. A helical electromagnetic interference shielding (HES) structure, comprising: a folded state; an unfolded state; a pair of rectangular EMI shielding panels that each comprise a first end and a second end; are affixed orthogonal to each other at their first ends; successively overlap each other to thereby form a plurality of interconnected EMI shielding planes; wherein each rectangular EMI

shielding panel comprises a substrate and a conductive layer; the substrate comprises a rectangular shape and one or more of a film and a fabric; the conductive layer is positioned proximate to the substrate and comprises a conductive composition; the HES structure unfolds to the unfolded state and folds to the folded state along its center axis; the conductive composition comprises a polymer and fully exfoliated single sheets of graphene that are present as a three-dimensional percolated network within the polymer; and the plurality of interconnected EMI shielding planes are each angularly offset from each other about the center axis to form a helical structure when in the unfolded state.

**2.** The HES structure of claim 1, wherein the plurality of interconnected EMI shielding planes each axially rotate about the center axis when the HES structure transitions between the folded state and the unfolded state.

**3.** The HES structure of claim 2, wherein adjacent EMI shielding planes comprise an angular offset positioned therebetween; the angular offset increases as distance between the adjacent EMI shielding planes increases; and decreases as distance between the adjacent EMI shielding planes decreases.

**4.** The HES structure of claim 3, wherein the plurality of interconnected EMI shielding planes comprise a top end and a bottom end; are rigidly affixed to a first rigid panel via the top end; are rotatably affixed to a second rigid panel via the bottom end; and are longitudinally positioned side by side to a second plurality of interconnected EMI shielding planes.

**5.** The HES structure of claim 4, wherein each rectangular EMI shielding panel comprises an encapsulating layer; the conductive layer is positioned proximate to a side of the substrate; the encapsulating layer encapsulates the rectangular EMI shielding panel; comprises one or more of a second film, a second fabric, and a second polymer; and is positioned proximate to one or more of the conductive layer and the substrate.

**6.** The HES structure of claim 5, wherein each rectangular EMI shielding panel comprises a metallic layer; the metallic layer comprises one or more of a metallic film, a metallic screen, and a metallic fabric; and is positioned proximate to one or more of the conductive layer, the substrate, and the encapsulating layer.

**7.** An electromagnetic interference (“EMI”) shielding construct, comprising: a plurality of the HES structures of claim 1 longitudinally arranged in rows and columns.

**8.** The EMI shielding construct of claim 7 in the form of a block, a wall, a curtain, or a divider.

**9.** The EMI shielding construct of claim 7 in the form of a panel, a liner, or a barrier.

**10.** The EMI shielding construct of claim 7, further comprising: a shell; and wherein the plurality of the HES structures is positioned within the shell.

**11.** An electromagnetic interference (“EMI”) shielding structure, comprising: a plurality of the EMI shielding constructs of claim 7.

**12.** The EMI shielding structure of claim 11 in the form of a room.

**13.** The EMI shielding structure of claim 11 in the form of a building.

**14.** The EMI shielding structure of claim 11 in the form of a structure comprising: a roof; and a plurality of walls.

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